Advanced Descriptive Statistics

★ Correlation :-

A statistical measure that describes the relationship b/w two variables / column.

Type of Correlation :-

1. Positive Correlation
2. Negative Correlation
3. No / Zero Correlation

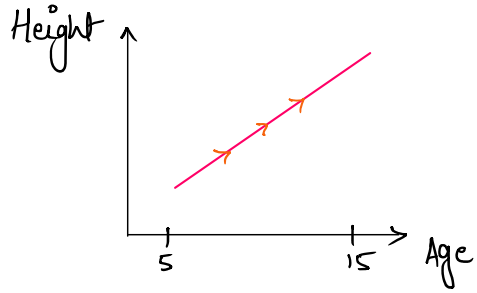
It is calculated by a coefficient known as coefficient of correlation

It is represented by 'r'.

Range → -1 to +1

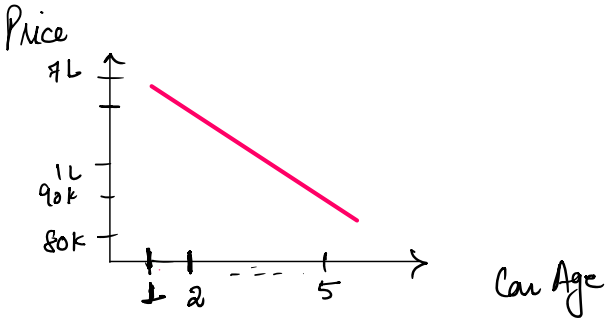
Positive Correlation :- When the value of one variable increases, the value of other variable also increases.

ex-



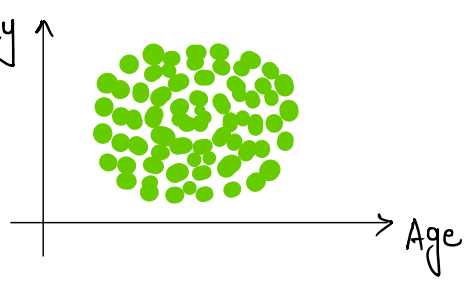
Negative Correlation :- When the value of one variable increases, the value of other variable decreases

ex-



Zero Correlation :- When there is no relationship or no effect

ex-



- Correlation is calculated using pandas dataframe.
- The strength of correlation tells us how strongly the two columns are related to each other.

Comparison:

Values of r	Type of Cor.
+1 or −1	Perfect Correlation
+0.7 to +0.95 or −0.7 to −0.95	Strong Correlation
+0.3 to +0.6 or −0.3 to −0.6	Moderate Cor.
0 to +0.2 or 0 to −0.2	Weak or No Correlation

★ Covariance :-

It tells us how one variable varies with respect to other variable

It also shows the relationship b/w the two variables

Covariance vs. Correlation

Covariance only tells us the direction (+ve/-ve) but not the strength b/w the two variables whereas can measure both the direction & strength

★ Skewness :-

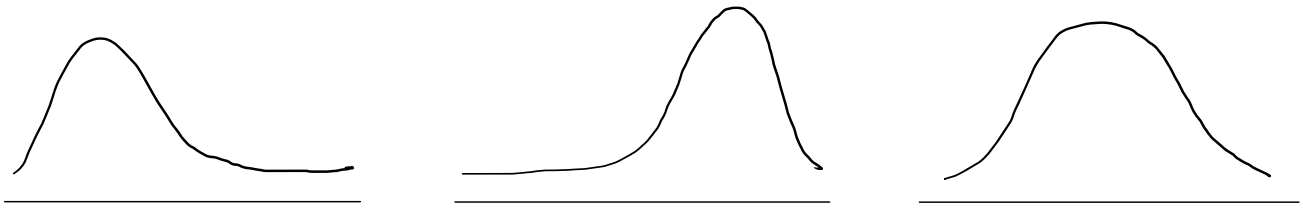
It tells us how the data is distributed

↓

shape / how does the data looks like?

Types of Skewed data :-

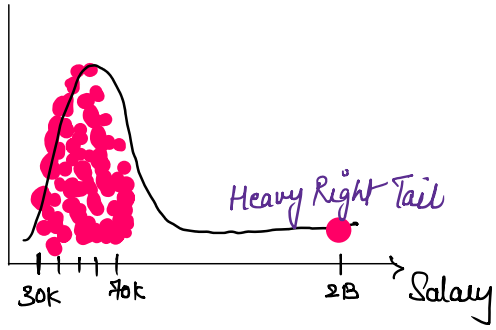
1. Positive Skewed (Right Skewed)
2. Negative Skewed (Left ")
3. Zero Skewed (Symmetric)



Right Skewed | Left Skewed | Zero Skewed

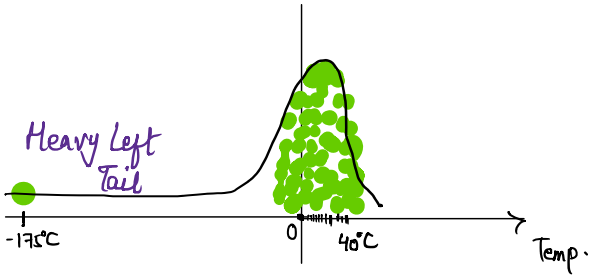
Ex-

Name	Exp.	Qual.	Sal.
●			70k
●			30k
●			38k
●			59k
●			2B ⇝ outlier
●			61k
●			40k

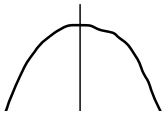


Ex-

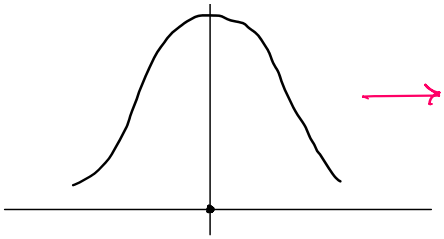
City	Pop.	Temp.
		32°C
		15°C
		-175°C ⇝ outlier
		20°C
		29°C
		19°C



Ex- Zero Skewed



→ Symmetric graph



→ symmetric graph

★ Kurtosis :-

It tells us that how heavy or light the tails are

Types :-

① High Kurtosis ⇝ Heavy Tails, more outliers

② Low Kurtosis ⇝ Lighter Tails, few outliers

③ Normal Kurtosis ⇝ No Tail, no outlier